REVA University • School of Computing

AIML-401: Deep Learning — Master Revision Notes

Topic: Backpropagation Algorithm & Multivariable Calculus Foundations

Candidate ID: STU-807 (Section B)	**Target Exam:** Midterm IA-2 & SEE
Curriculum Unit: Unit 3: Deep Neural Networks	**Generated:** Nov 02, 2026 - 17:40
Faculty Source: FAC_AIML_01 (backprop_lecture_03.pdf)	**Status:** High-Yield Mastery Guide

1. Core Concept & Theoretical Definition

Backpropagation is an exact algorithmic implementation of the reverse-mode automatic differentiation technique applied to composite loss functions in multi-layer artificial neural networks. Its foundational objective is to calculate the partial derivative of the scalar loss L with respect to every learnable parameter (weight w and bias b) efficiently in O(N) operations per sample.

2. Mathematical Formulation & Multivariable Chain Rule

For an output neuron k with net weighted input $z_k = \sum w_{jk} a_j + b_k$ and activation $y_k = \sigma(z_k)$, the error gradient flows recursively:

$$\text{Weight Gradient: } \partial L / \partial w_{jk} = (\partial L / \partial y_k) \cdot \sigma'(z_k) \cdot a_j = \delta_k \cdot a_j$$
$$\text{Hidden Layer Gradient: } \delta_j = (\sum \delta_k w_{jk}) \cdot \sigma'(z_j)$$

Key Takeaway for Exams: *Notice that the error δ at hidden neuron j is a linear combination of all downstream errors δ_k weighted by the connecting forward weights w_{jk}, scaled by local activation derivative $\sigma'(z_j)$.*

3. Step-by-Step Backpropagation Algorithm

Step	Phase	Mathematical Operation	Computational Complexity
1	Forward Pass	Compute net activations z = W a + b and output y = f(z) across all layers	O(W) Matrix Multiply
2	Loss Evaluation	Calculate total scalar error: L = (1/2) * sum((y_pred - y_true)^2)	O(K) Vector Norm
3	Backward Pass	Calculate output delta: delta_k = (y_pred - y_true) * f'(z_k), propagate back	O(W) Transposed Multiply
4	Weight Update	w_new = w_old - eta * (delta * a_prev^T) via Gradient Descent	O(W) Elementwise Add

4. Faculty Notes Citations & Exam Focus Areas

Retrieved Excerpt from Faculty Slide Deck (backprop_lecture_03.pdf - Slide 14):
'Partial derivatives of composite functions must strictly adhere to the multivariable chain rule. Common exam mistakes include forgetting the activation derivative term $\sigma'(z)$ and failing to sum across multiple fan-out paths from a hidden node.'

5. Predicted High-Yield Question for IA-2 / SEE

Question (8 Marks): Derive the backpropagation weight update equation for a hidden layer in a 3-layer MLP using sigmoid activations and MSE loss.
Scoring Blueprint:

- 2 Marks: Clear diagram of forward signal flow and backward error flow.
- 3 Marks: Correct expansion of partial derivatives using chain rule: $\partial L/\partial w = \partial L/\partial z \bullet \partial z/\partial w$.
- 3 Marks: Vectorized final update formula: $W^{(l)} \leftarrow W^{(l)} - \eta\, \delta^{(l)} (a^{(l-1)})^T$.